

Assignment Title: - No. 6

Create Online appointment booking application / E-commerce portal for used items sales like car, bike etc. or similar type application using Node.js and Express use any NoSQL data set like MongoDB if required.

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Objectives: -

- To develop a responsive, full-stack E-commerce platform for fashion enthusiasts.
- To implement secure, role-based authentication (Admin vs. User) using JWT.
- To manage a dynamic product catalog with CRUD operations for administrators.
- To integrate a third-party payment gateway (Razorpay) for secure financial transactions.
- To provide a seamless user experience with search, filtering, and real-time cart updates.

Outcomes: -

- Data Persistence: Successful storage and retrieval of users, products, and orders using MongoDB.

- Security: Prevention of unauthorized access to the admin dashboard and user cart.
- Interactivity: A dynamic React-based UI that updates state (like cart counts) without page refreshes.
- Financial Integration: Successfully completing the payment lifecycle from checkout to payment confirmation.

Theory Concept in brief: -

- MERN Architecture: A 3-tier architecture consisting of a React frontend, Node/Express backend, and MongoDB database.
- RESTful APIs: Communication between the frontend and backend using standard HTTP methods (GET, POST, PUT, DELETE).
- State Management: Using React Hooks (useState, useEffect, useContext) to manage global app data like the shopping cart.
- Authentication & Authorization: Using JWTs to verify identity and Middleware to protect specific routes based on user roles.

Technologies Used: -

<u>Layer</u>	<u>Technology</u>
<u>Frontend</u>	<u>React.js, Bootstrap 5 (CDN), CSS3, HTML5</u>
<u>Backend</u>	<u>Node.js, Express.js</u>
<u>Database</u>	<u>MongoDB Atlas</u>
<u>Security</u>	<u>JWT, Bcrypt.js (password hashing)</u>
<u>Payment</u>	<u>Razorpay API</u>
<u>Hosting/Tools</u>	<u>Postman (API Testing), Git</u>

Dataset Used: -

- The application uses a Non-Relational (BSON) dataset stored in MongoDB.
- Users Collection: Contains name, email, password (hashed), and role (Admin/User).
- Products Collection: Contains title, description, price, images (URLs), category, sizes[], and stockQuantity.

- Orders Collection: Contains userID, productItems[], totalAmount, paymentStatus, and shippingStatus.

Functional Description: -

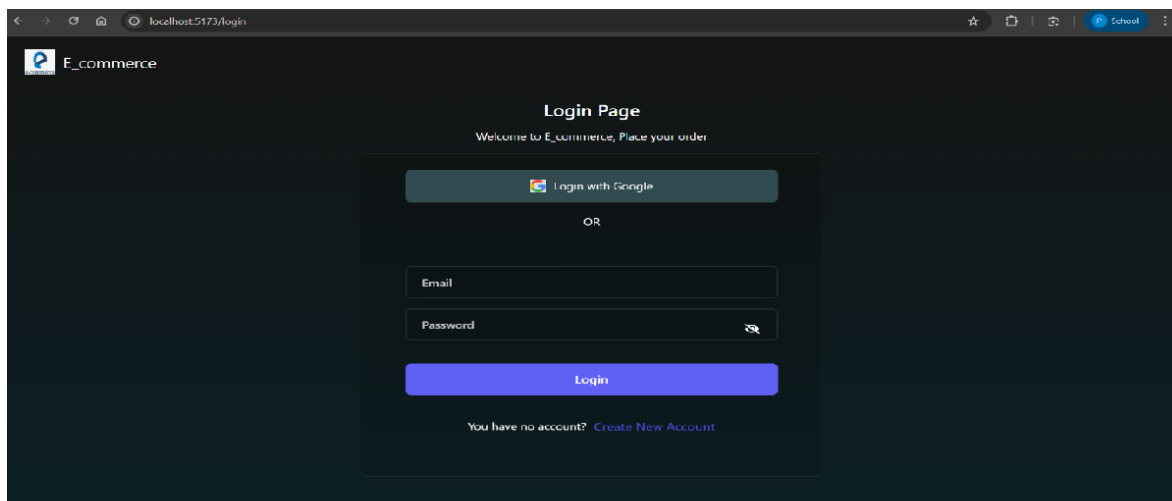
- Authentication Module
- Sign Up/Login: Users enter credentials; the backend verifies them and sends a JWT stored in the browser (LocalStorage/Cookies).
- Role Verification: The frontend checks the JWT payload to decide whether to show the "Admin Dashboard" link or the "Cart" icon.
- Product Discovery (User Side)
- Search & Filter: Users can search for specific items. The React frontend filters the product list in real-time.
- Similar Products: When viewing a product, the app fetches other products from the same category using a Category-ID query.
- Cart & Checkout
- Cart Logic: Adding an item triggers a state update. The cart persists even after refresh if stored in the database.

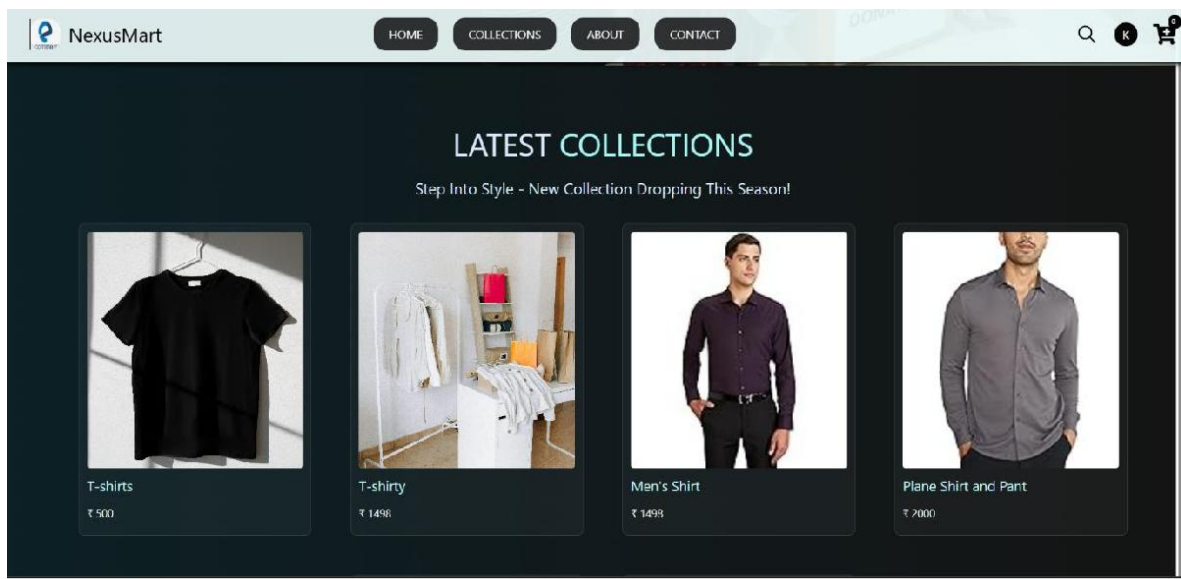
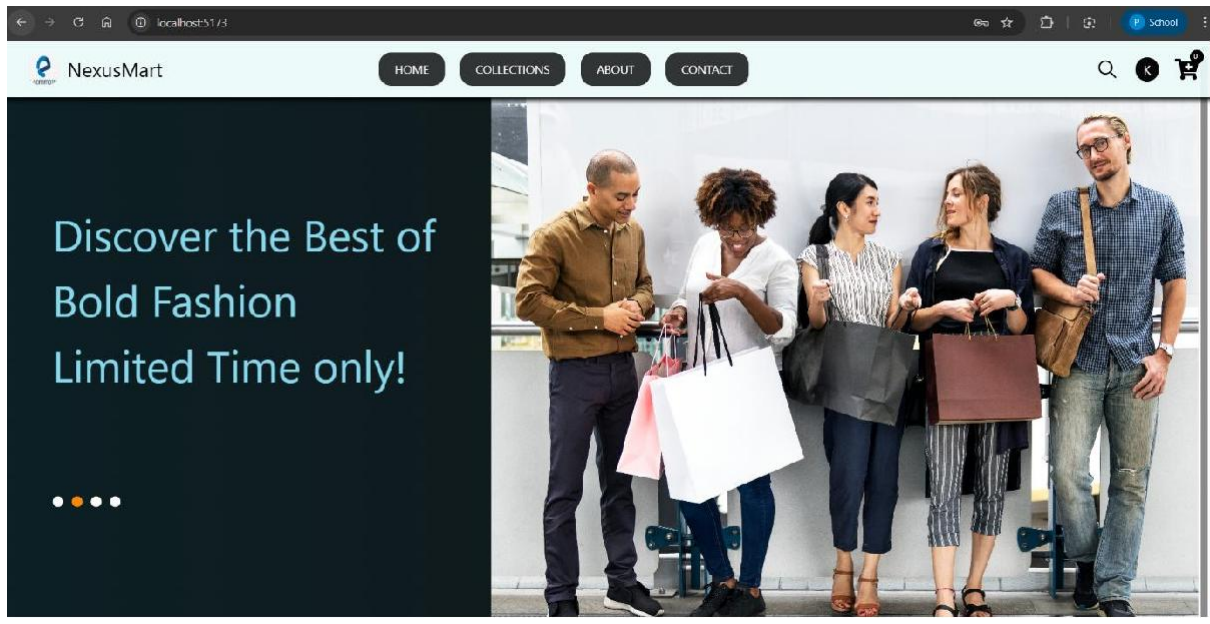
- **Razorpay Integration:** When the user clicks "Checkout," the backend generates an orderID via Razorpay. The frontend opens the Razorpay popup. Upon success, the payment details are saved.
- **Admin Dashboard**
- **Inventory Management:** Admin can fill a form to upload a new fashion item.
- **Order Tracking:** Admin sees a list of all user purchases and can toggle the status from "Pending" to "Shipped."

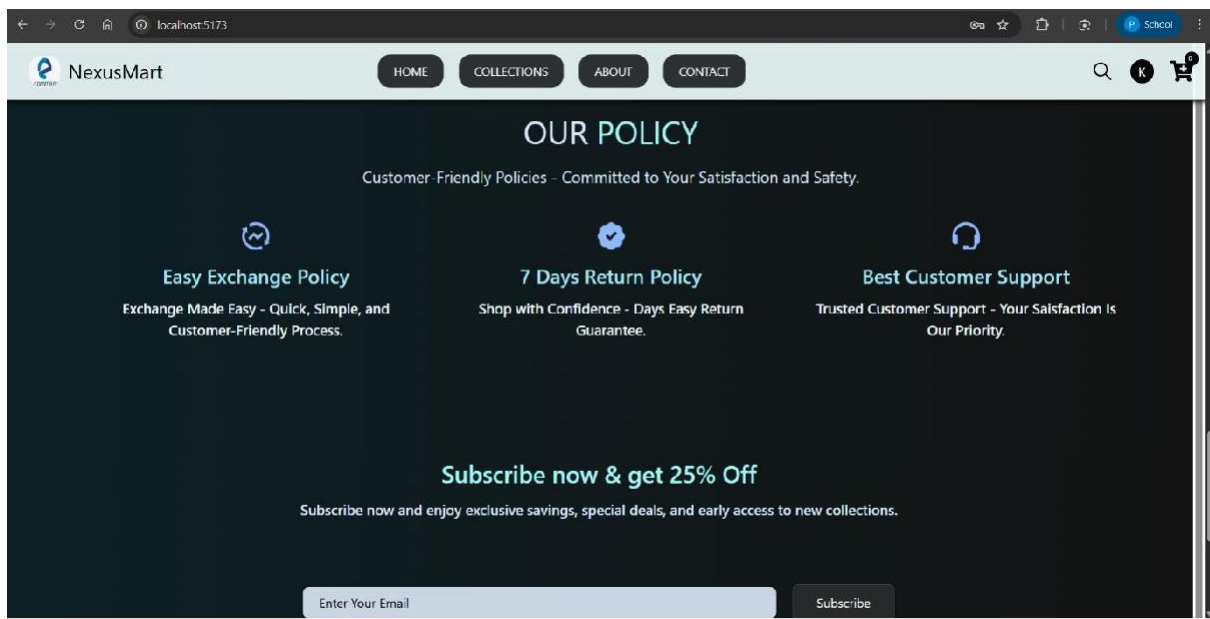
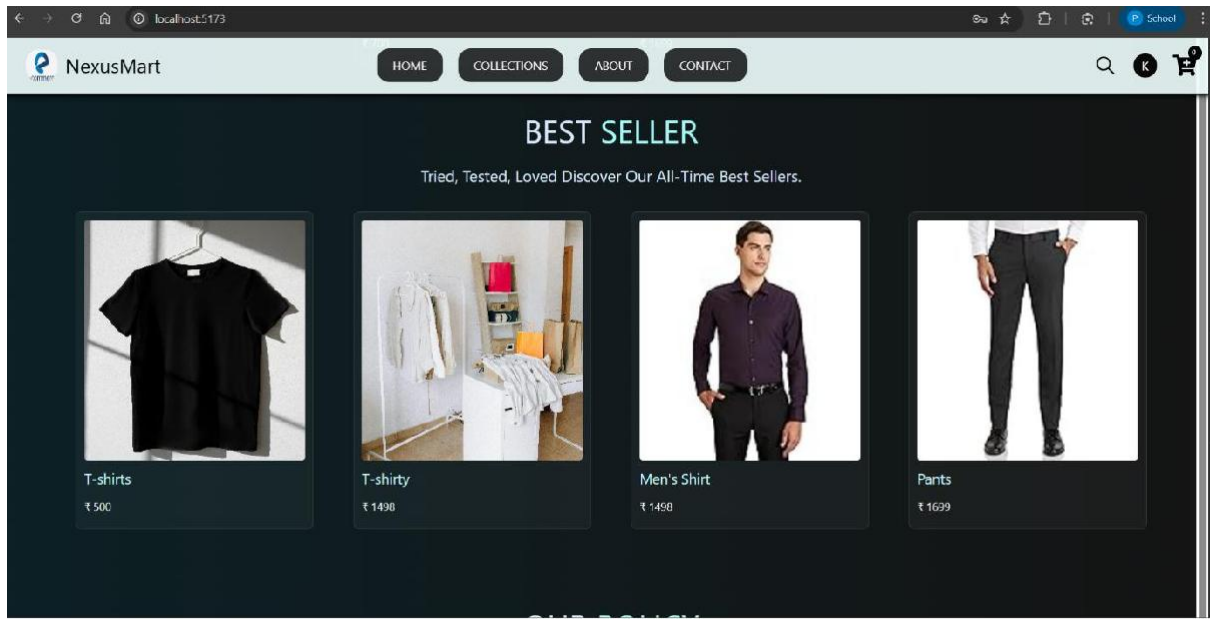
Program Code with output Screenshot: -

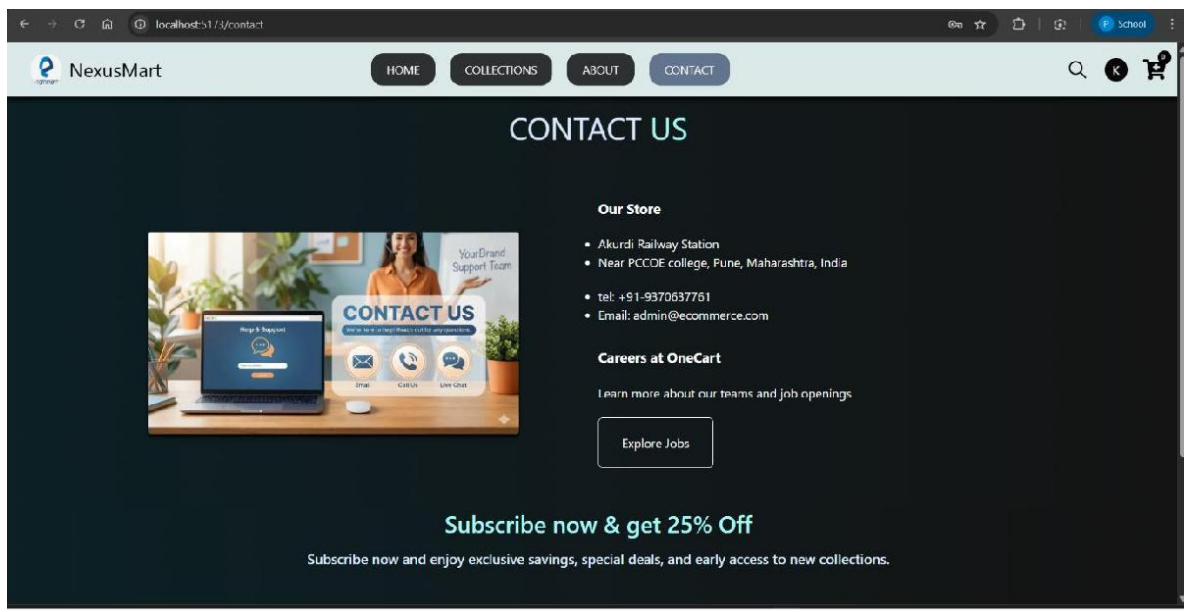
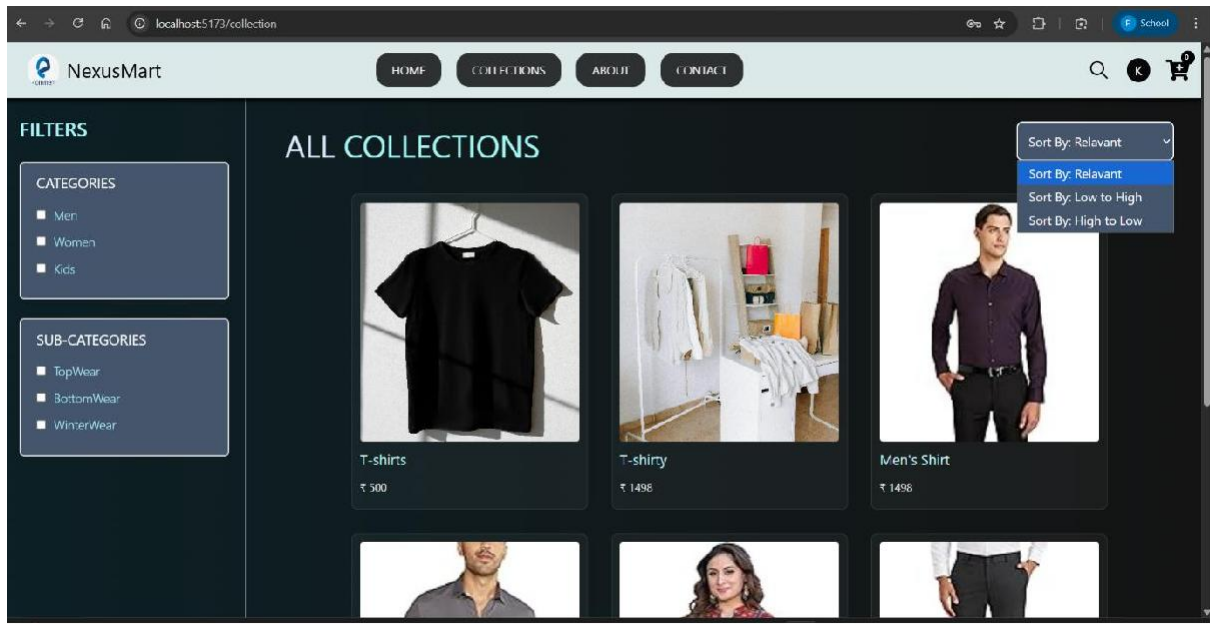
Program code is attached with zip File :-
E_commerce Website.zip

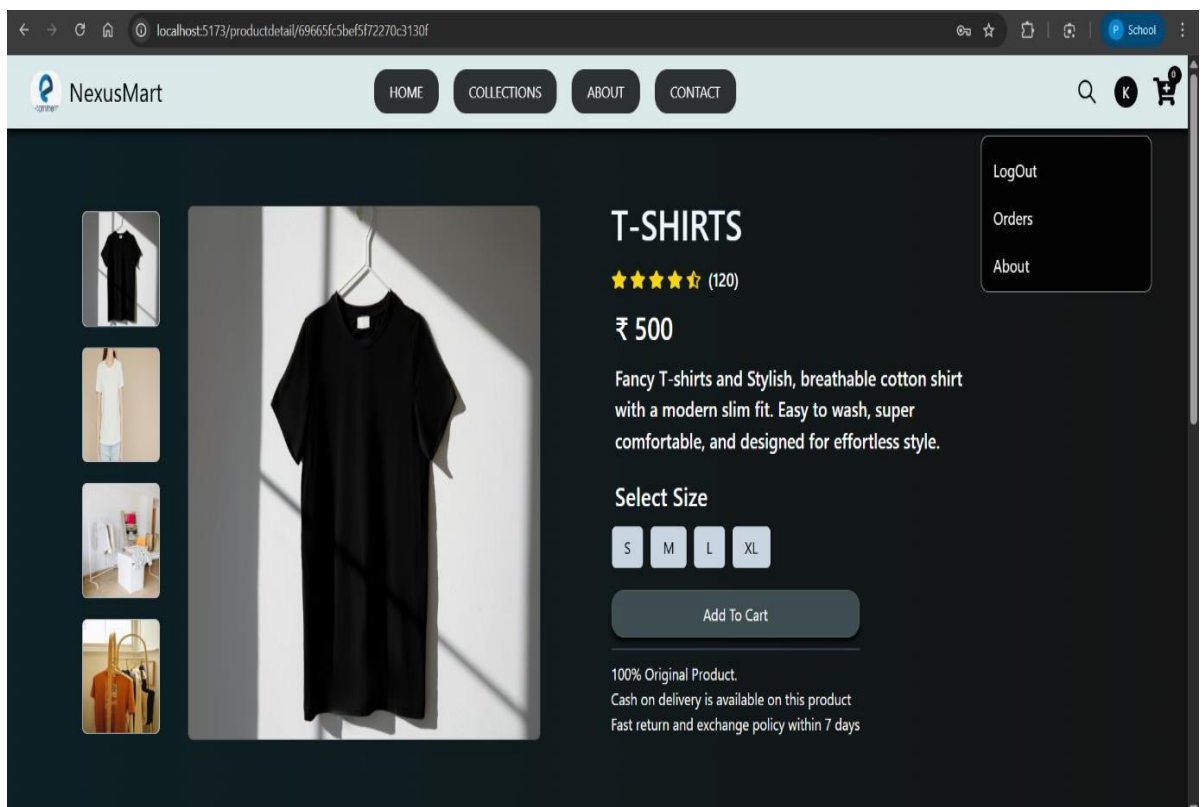
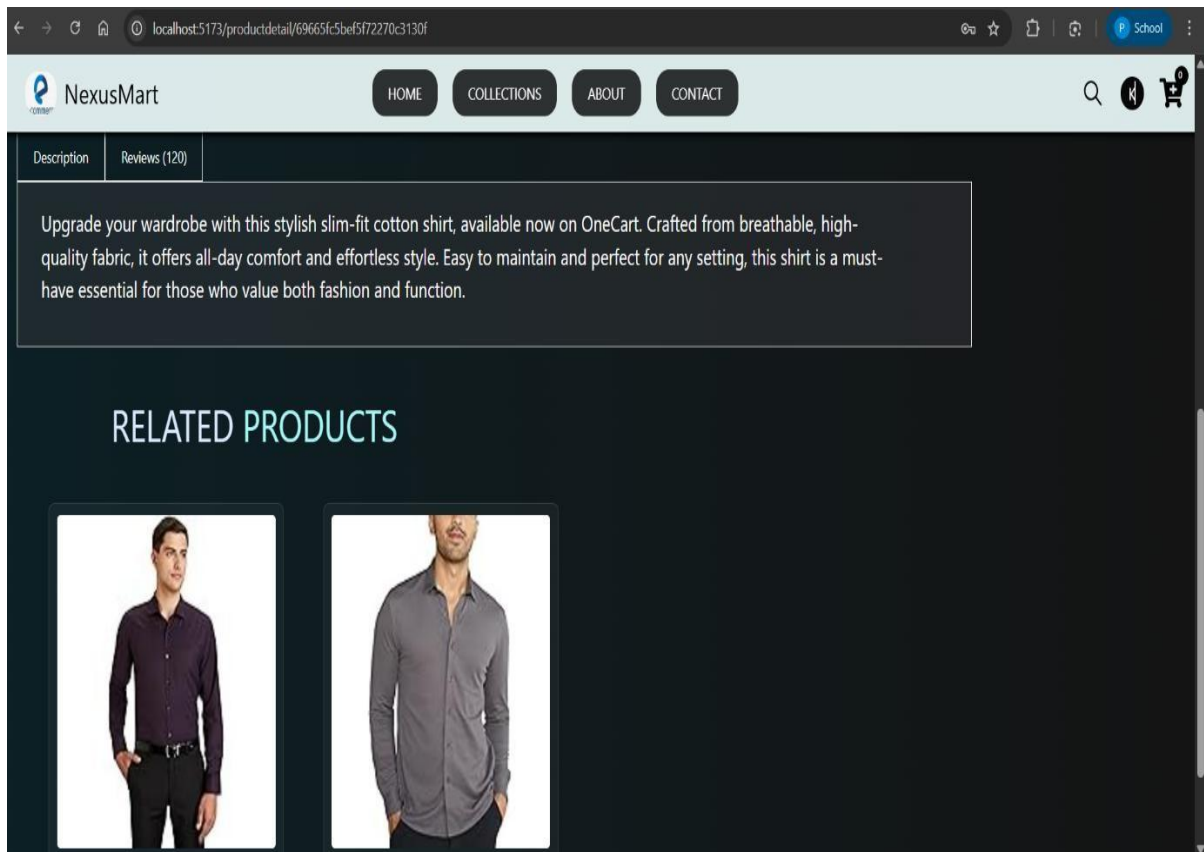
Program Output: -

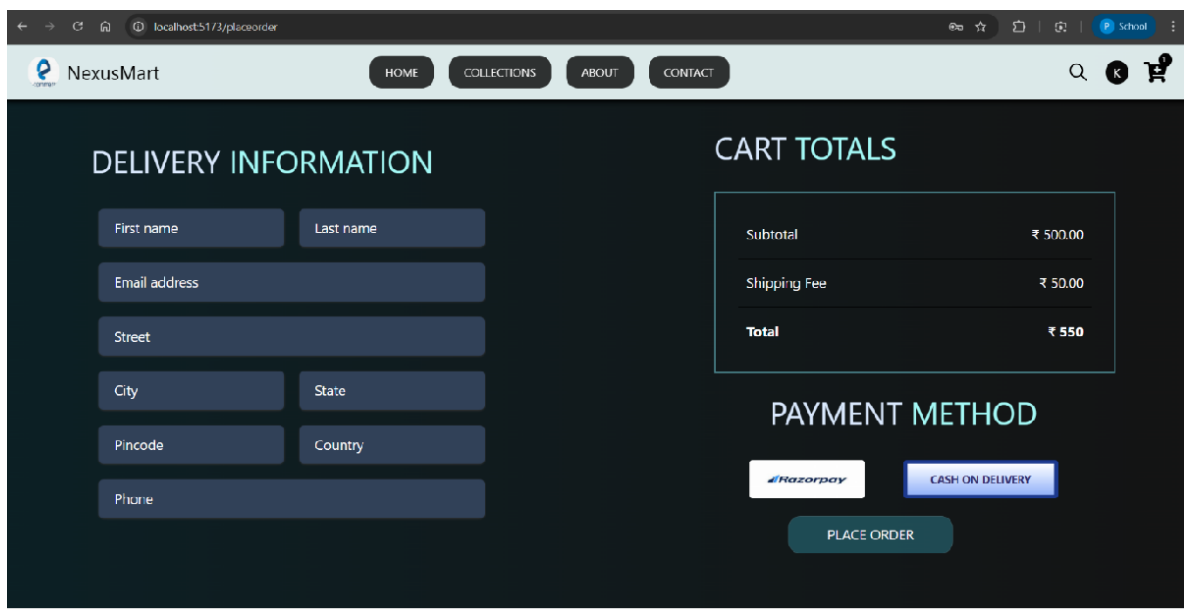
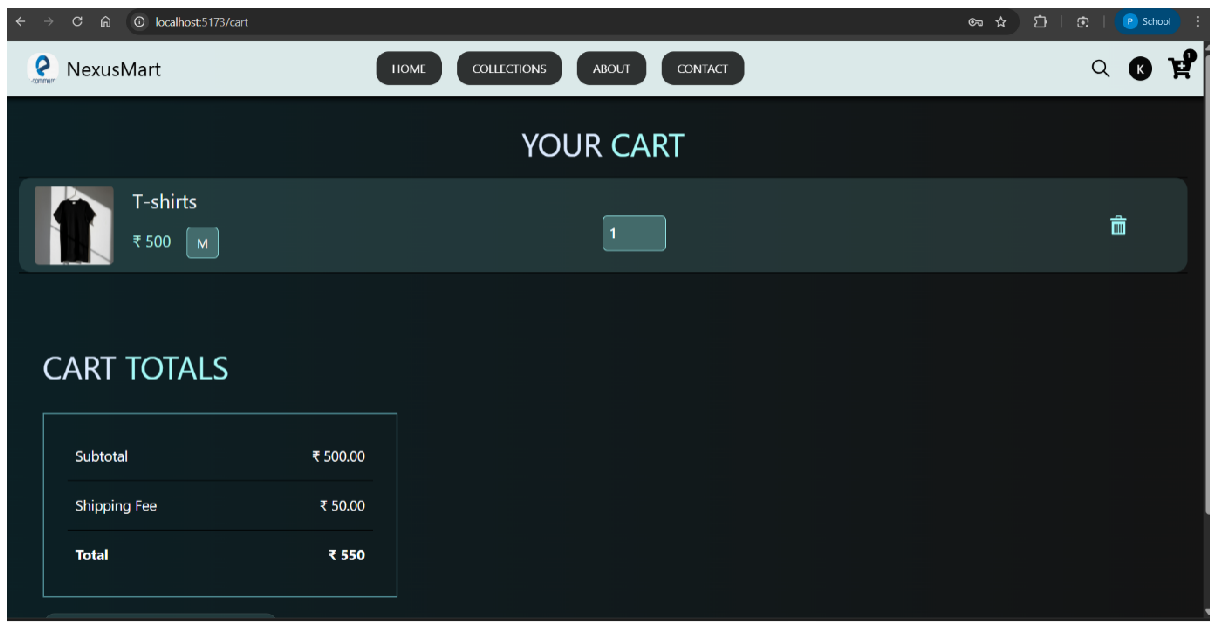












localhost:5174/add

E_commerce

LogOut

Add Product Page

Upload Images

Product Name

Type here

Product Description

Type here

Product Category **Sub-Category**

Admin Side: -

localhost:5174/add

E_commerce

LogOut

Add Product Page

Product Category **Sub-Category**

Men TopWear

Product Price

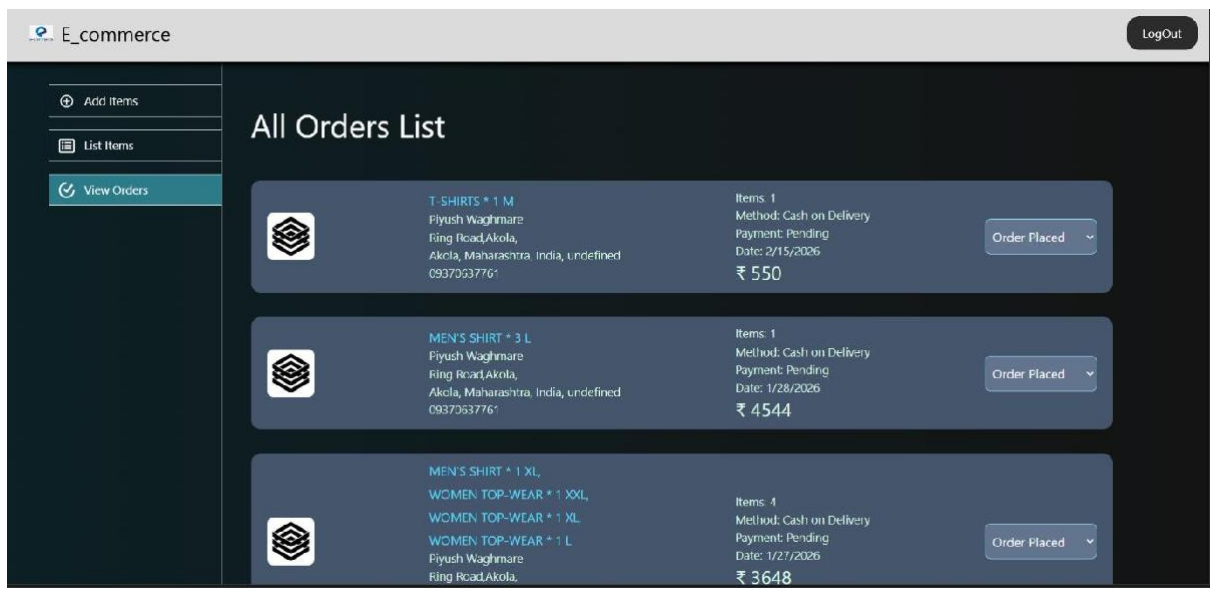
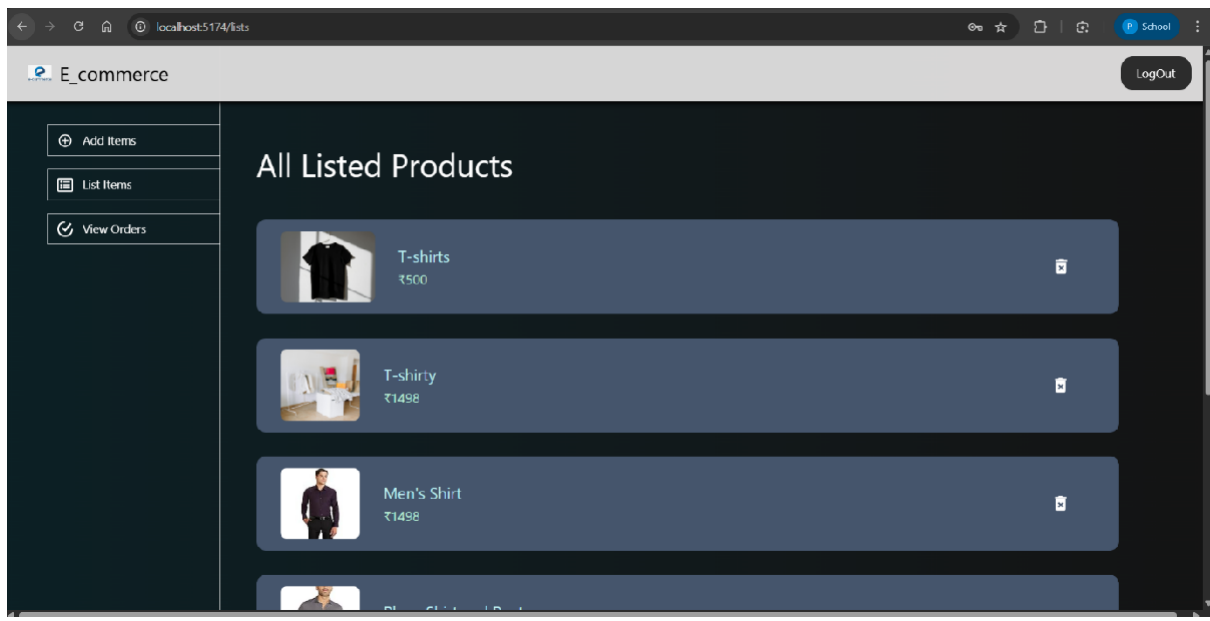
₹ 2000

Product Size

S M L XL XXL

☐ Add to BestSeller

Add Product



Conclusion: -

The "Elite Couture" MERN application successfully demonstrates a scalable approach to modern web development. By combining the flexibility of MongoDB with the reactive nature of React, the store provides a

high-performance shopping experience. The inclusion of JWT and Razorpay ensures that the application is not just a UI prototype, but a production-ready, secure platform.